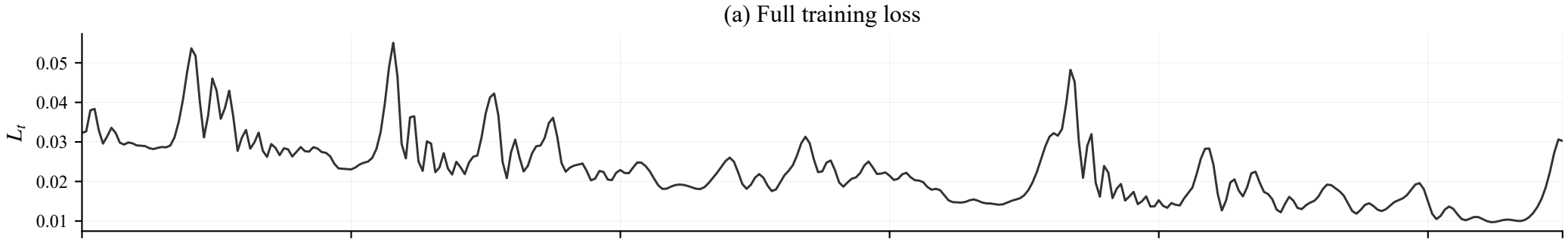
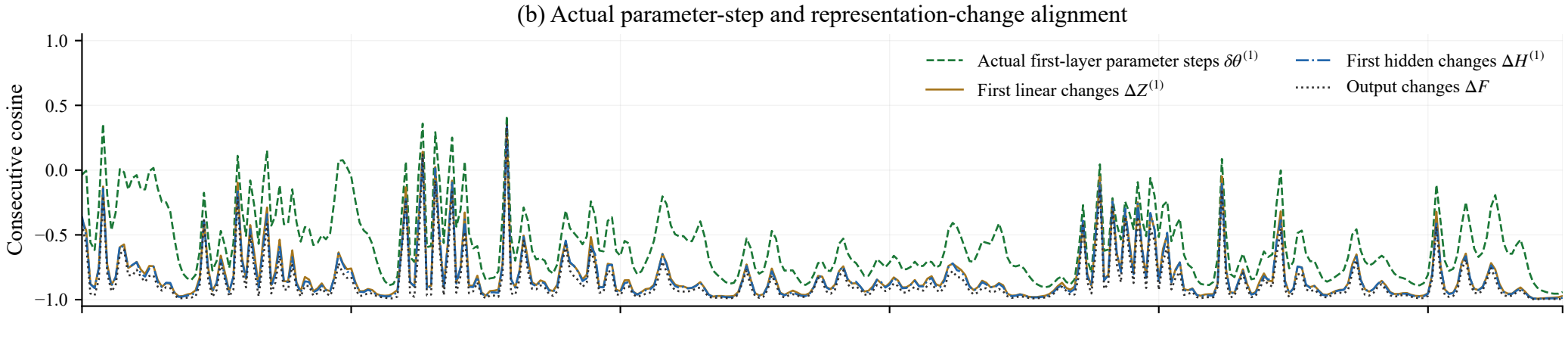
CIFAR-10 MLP: fixed images, changing first-layer projection

$\Delta Z_t^{(1)} = X\Delta W_{1,t}^T + \Delta b_{1,t}$; $\Delta H_t^{(1)} = S_t \odot \Delta Z_t^{(1)}$, with measured forward-roundoff retained.

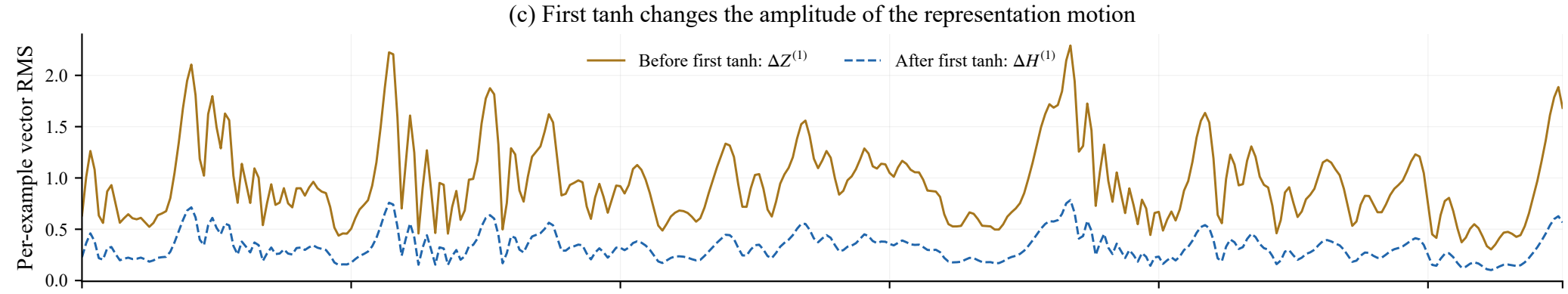
(a) Full training loss



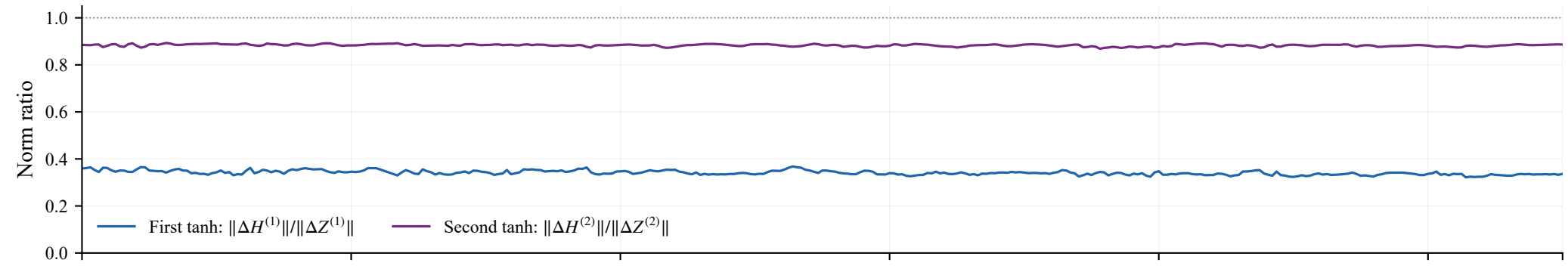
(b) Actual parameter-step and representation-change alignment



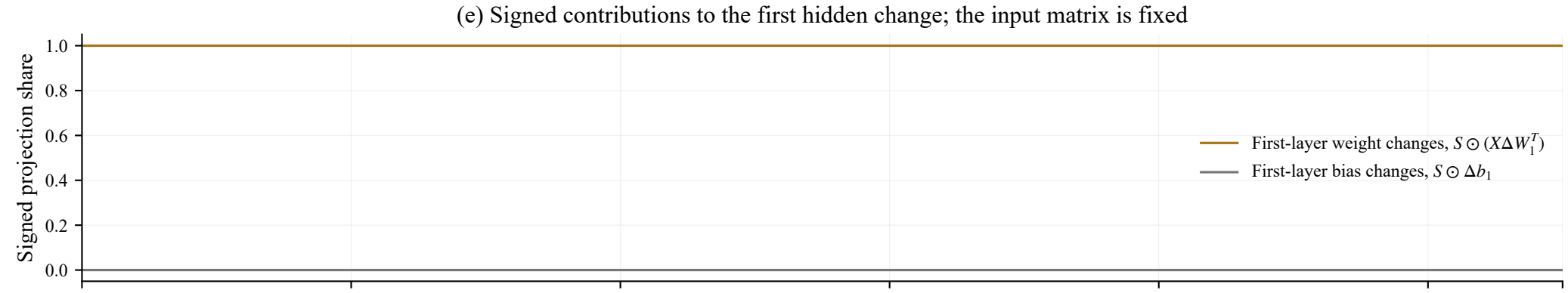
(c) First tanh changes the amplitude of the representation motion



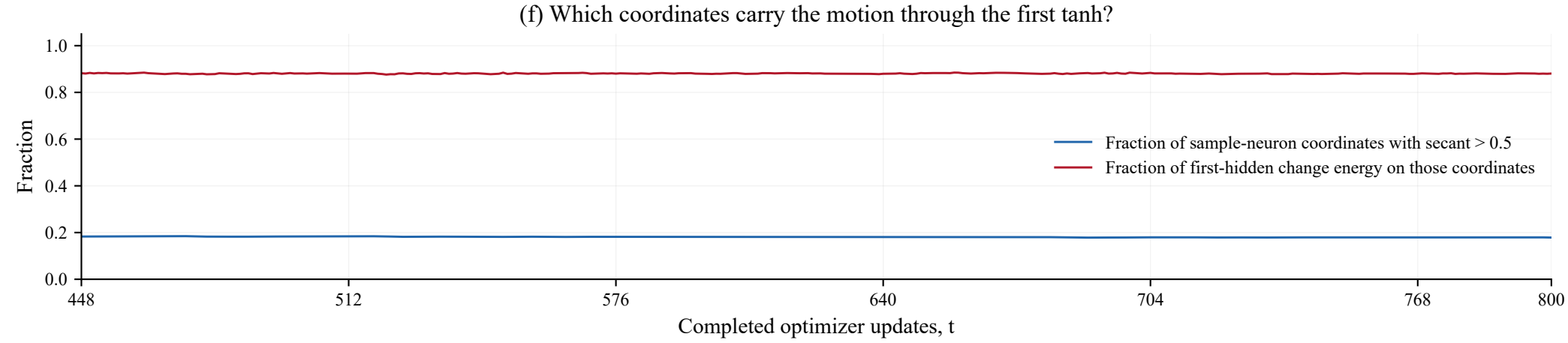
(d) Measured finite-displacement gain of both tanh layers



(e) Signed contributions to the first hidden change; the input matrix is fixed



(f) Which coordinates carry the motion through the first tanh?



Actual parameter increments are measured from successive weights and biases; they are not raw gradients or gradient differences.

The secant is the elementwise finite tanh slope. Threshold 0.5 summarizes transmission strength, not a binary neuron state.

These observations locate the earliest changing representation; they do not identify the unique cause of the optimizer oscillation.